SYMMETRY

How to Gauge a Discrete Symmetry?

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Abstract. This note is about gauging a discrete symmetry. We will first review what gauging is in a continuous case and then try to define a counterpart for a discrete symmetry. Finally, we will introduce concrete example in Virasoro Minimal Models. For generalized symmetry part we mainly follow [1], for CFT part one can learn about the modular invariant minimal model in [2].

Contents

Gauging is the case when we couple a QFT to a gauge field (often dynamical, but some people may also call background gauge field as gauging). Apart from its importance of studying interaction between QFTs, gauging is also a powerful tool to study the symmetry of a QFT. The core idea is that it may serve as a test to probe whether the symmetry really exist or there's an anomaly.

1 How to Gauge a Continuous Symmetry?

First we review some basics of coupling a QFT to a gauge field. Of course, we can just do the minimal coupling and be happy with it. However, to make it more general, we should not assume we always have a conserved current or even a classical lagrangian description.

1.1 What is a Symmetry?

To characterize a symmetry in QFT, we can use the conserved current or the conserved charge. However, these are objects unique to the continuous symmetry and expects the theory to have a classical counterpart. To make thing general, we should think about another object **the symmetry generator**.

1.1.1 Classical Definition

We first take an example of a classical field theory with a U(1) symmetry (say a complex scalar field). Noether's theorem tells us that there is a conserved current j satisfying the conservation equation:

$$d * j = 0 \tag{1}$$

And we can define a conserved charge by integrating the current over a spatial slice:

$$Q = \oint_{\Sigma} * j \tag{2}$$

where Σ is a spatial slice. Now we can try to integrate the current over a codimension-1 manifold M_{d-1}, which is not necessarily a spatial slice, and define a charge as:

$$Q(M_{d-1}) = \oint_{M_{d-1}} * j \tag{3}$$

and a related classical quantity we called the **symmetry generator**:

$$U(M_{d-1}) = \exp(i\alpha Q(M_{d-1})) \tag{4}$$

One can prove that these objects are topological, which means that they are invariant under continuous deformation of the manifold M_{d-1}, as long as we don't cross any strange stuffs. This can be proven with the stokes theorem:

$$Q(M_{d-1}) = \oint_{M_{d-1}} * j = \oint_{M'_{d-1}} * j + \oint_{B_d} d * j = Q(M'_{d-1}) \tag{5}$$

where B_d is the manifold connecting M_{d-1} and M'_{d-1} satisfying:

$$\partial B_d = M_{d-1} \sqcup M'_{d-1} \tag{6}$$

1.1.2 Quantum Interpretation

Now we generalize the classical object we define above to a quantum one.

> **Note**
>
> Notice that the world is NOT classical, it is definitely true that we can have some quantum object without classical counterparts. Namingly can't be derived from quantizing some classical object.
>
> Thus, we don't derive a quantum theory from a classical one, but use some quantization scheme and results to guess what properties should a quantum symmetry generator have.

The properties of **quantum symmetry generators** (and justifications) are:

1. They are **Topological Operators**. This can be justified by the classical conservation equation, plugging them into a path integral won't change this fact. Moreover, as a quantum

theory, in the canonical picture if we want to define these symmetry generators as line operators, they have to commute with the Energy-Momentum tensor, which generates the spacetime translation.

2. If inserted in a time slice, they act as an operators on the Hilbert space in the canonical formalism. This can be justified by analoging the ordinary symmetry charge defined on a time slice.
3. If inserted in vetically in the path integral, they behave as a defect, which twisted the Hilbert Space. This can't be justified straightforwardly, but a few lines of calculation in the path integral formalism:

YL: [finish this part later]

(2 and 3 properties are called the operator/defect principles)

4. When going through local operators, they act as a symmetry transformation on the local operators. This can be justified by the Ward identity.

YL: [finish this part later]

1.1.3 A Slogan for Symmetry

At this point it is now wise to introduce a slogan to characterize symmetry beyond the current/charge picture, which limits in continous symmetry with a classical lagrangian description.

> **Definition 1.1** **Symmetry-Defect Slogan**
>
> A symmetry should be characterized by the existing of a topological operator (Symmetry Generators), having above properties and many many constraints.

This is a powerful generalization of the notion of symmetry, which eventually we will see that it can be applied to continuous symmetries, discrete symmetries, and even symmetries without a inverse, called non-invertible symmetries.

1.2 What is a Gauge Field?

Now we have a grasp of what is a symmetry, we turn to the other question of gauging: what is a gauge field?

Naively, we define a gauge field as a Lie algebra valued one form field:

$$A = A_\mu dx^\mu \text{ where } A_\mu \in \mathfrak{g} \tag{7}$$

However, this is generally not the case. A gauge field may not be globally defined.

A classical and experimental example is the AB effect of U(1) gauge field, where we know that:

$$\exp\left(i \oint_C A \, \mathrm{d}l\right) \tag{8}$$

and is exactly an experimental observable. Thus we need to rethink the definition of gauge field. The critical point is that a gauge field may not be defined globally, but only on patches, while field on different patches are related by a suitable gauge transformation.

Moreover, from demanding consistency in AB phase and a globally well defined curvature (indeed we measure electric and magnetic fields and don't see any singularity), we need to have a quantization condition on the flux of the gauge field:

$$\oint_S F \in 2\pi\mathbb{Z} \tag{9}$$

1.2.1 Mathematical Characterization

A mathematical framework, namingly **fibre bundle** is developed to characterize all these features of gauge field (which I wish to introduce in another note). Now to make things simple, we try to construct a gauge field step by steps:

- **Prepare Patches of Principle G Bundle**:

1. Given a manifold M_d choose a set of open over $\{U_i\}$
2. On each patch U_i we "choose a trivialization of the Pinciple G-bundle". This is kinds of equivalent to another fancy name, which is choosing a **section** of the bundle, which is a map from the base space to the total space of the bundle:

$$s_i : U_i \to P \tag{10}$$

well we can just understand as mapping each point in the patch to a group element and its only defined locally. Physically this is equivalent to choosing a gauge, as discussed before, we can't choose a global gauge, thus we can only choose a local gauge on each patch.

3. On the overlap of two patches U_i and U_j, we have a transition function:

$$g_{ij} : U_i \cap U_j \to G \tag{11}$$

this relates the two trivialization on the two patches, and we have:

$$s_i = g_{ij} s_j \tag{12}$$

4. On a triple overlap of three patches U_i, U_j and U_k, we have a consistency condition:

$$g_{ij} g_{jk} g_{ki} = 1 \tag{13}$$

- **Construct Gauge Field**:

gauge field can be understood as a connection on the principle G-bundle after a pullback by the section we choose. But we can just understand it as a well defined Lie algebra valued one form on each patch:

$$A^{(i)} \in \Omega^1(U_i) \otimes \mathfrak{g} \tag{14}$$

and on the overlap of two patches, we have a gauge transformation:

$$A^{(j)} = g_{ij}^{-1} A^{(i)} g_{ij} + g_{ij}^{-1} d g_{ij} \tag{15}$$

Two important objects of the principle bundle are the curvature and the holonomy of the gauge field, which are defined as:

$$F^{(i)} = dA^{(i)} + A^{(i)} \wedge A^{(i)} \tag{16}$$

$$\mathrm{Hol}(C) = \exp\left(i \oint_C A \right) \tag{17}$$

physically, they correspond to the field strength and the AB phase of U(1) gauge field.

1.2.2 Quantization of Flux

With suitable consistent mathematical construction of gauge field from fibre bundles, we can see that quantized flux arises naturally from mathematical properties of fibre bundles.

- Quantization of U(1) flux is related to the first Chern class of the U(1) bundle, which is an integer.

$$c_1 = \frac{1}{2\pi} \oint_S F \in \mathbb{Z} \tag{18}$$

- Quantization of SU(N) non-abelian flux is related to the second Chern class of the non-abelian bundle, which is also an integer.

$$c_2 = \frac{1}{8\pi^2} \oint_S \mathrm{Tr}(F \wedge F) \in \mathbb{Z} \tag{19}$$

1.3 How to Couple to a Gauge Field?

Now we start with the true topic, Gauging. By gauging a symmetry, we mean coupling the theory to a gauge field and make it dynamical.

1.3.1 BG Gauge Field Insertion of Symmetry Generator

1.3.2 AB phase as going through Symmetry Generator

Now we consider local operators in the theory. Notice that the local charged operators are no longer gauge invariant observables but rather operators with Wilson line attached.

1.4 Anomaly

2 How to Gauge a Discrete Symmetry?

Bibliography

[1] J. Kaidi, "Introduction to Generalized Symmetries." Accessed: Apr. 13, 2026. [Online]. Available: http://arxiv.org/abs/2603.08798

[2] P. Di Francesco, P. Mathieu, and D. Sénéchal, *Conformal Field Theory*. in Graduate Texts in Contemporary Physics. New York, NY: Springer New York, 1997. doi: 10.1007/978-1-4612-2256-9.